

Part I: Perception

Raw Document Image

PDF / Scan / Photo

Layout Analysis

Tables, Figs, Headings
DocLayout-YOLO, DLAFormer

OCR

Text Recognition
PaddleOCR, GOT, MonkeyOCR

Structured Textx + Layout Info

JSON / Markdown / HTML

Part II: Understanding

Table Understanding

Structure → Modular → Verifiable
TableGPT2, Chain-of-Table

Text RAG

Naive → Graph → Agentic RAG
GraphRAG, GARLIC, ColPali

Multimodal VLM

OCR-based → OCR-free → Multi-page
Qwen2-VL, mPLUG-DocOwl2

Document QA / Answer

Exact / Abstractive / Evidence